Hospital Management System (HMS)

Complete Project Workflow & Module Structure

A complete end-to-end workflow for a modern, clear and user-friendly Hospital Management System. This document can be used as a reference for UI/UX design, system analysis, database planning and development.

1. System Overview

```
LOGIN
  ↓
ROLE DETECTION
  ↓
DASHBOARD
  ↓
Hospital Modules
  ├── Administration
  ├── Reception
  ├── Doctor
  ├── Nurse
  ├── Laboratory
  ├── Pharmacy
  ├── Billing & Payments
  └── Patient Portal
```

2. Main User Roles

1. Admin
2. Doctor
3. Nurse
4. Receptionist
5. Pharmacist
6. Laboratory Staff
7. Patient

3. Patient Registration Workflow

```
Patient arrives
      ↓
Receptionist
      ↓
Search Patient
      ↓
Does patient exist?
   ↙         ↘
 YES          NO
 ↓            ↓
Open Profile   Register Patient
 ↓            ↓
Update info    Generate Patient ID
      ↘       ↙
        Patient Profile
             ↓
        Book Appointment
```

4. Patient Information

- Patient ID
- Full name
- Gender
- Date of birth
- Phone

• Address
• Emergency contact
• Blood group
• Allergies
• Medical history

5. Appointment Workflow

Patient / Receptionist
↓
Book Appointment
↓
Select Department
↓
Select Doctor
↓
Select Date & Time
↓
Confirm Appointment
↓
Appointment Status = PENDING
↓
Doctor confirms
↓
Status = CONFIRMED
↓
Patient receives notification

6. Patient Arrival / Check-in

Patient arrives
↓
Receptionist searches Patient ID
↓
Today's appointment?
↙ ↘
YES NO
↓ ↓
Check-in Create/Book visit
↓ ↓
Generate Queue Number
↓
Patient waits
↓
Nurse calls patient

7. Nurse Workflow

Nurse Dashboard
↓
View Patient Queue
↓
Call Patient
↓
Open Patient Profile
↓
Record Vital Signs
↓
Blood Pressure
Temperature
Heart Rate
Weight
Oxygen Level
↓
Add Nursing Notes
↓
Send Patient to Doctor

8. Doctor Consultation Workflow

Doctor Dashboard
↓
Today's Patient Queue
↓
Select Patient
↓
View Medical History
↓
Review Vital Signs
↓
Consultation
↓
Doctor records:
- Symptoms
- Diagnosis
- Doctor Notes
- Treatment
- Follow-up

↓
Does patient need Lab Test?
↙ ↘
YES NO
↓ ↓
Lab Request Prescription
↓ ↓
Laboratory Pharmacy

9. Laboratory Workflow

Doctor requests test
↓
Lab receives request
↓
Lab Dashboard
↓
Patient arrives
↓
Collect Sample
↓
Status = SAMPLE COLLECTED
↓
Perform Test
↓
Enter Results
↓
Doctor reviews results
↓
Patient Medical Record updated

10. Pharmacy Workflow

Doctor creates prescription
↓
Pharmacy receives prescription
↓
Pharmacist checks medicine
↓
Medicine available?
↙ ↘
YES NO
↓ ↓
Dispense Out of Stock
Medicine ↓
↓ Notify Admin
Update Stock
↓
Prescription = COMPLETED
↓

Patient receives medicine

11. Billing & Payment Workflow

Registration
↓
Consultation
↓
Laboratory
↓
Pharmacy
↓
Other Services
↓
Generate Invoice
↓
Calculate Total
↓
Patient Payment
↓
Payment Status
↙ ↓ ↘
PAID PARTIAL PENDING
↓ ↓ ↓
Receipt Balance Reminder

12. Medical Records Workflow

PATIENT PROFILE
│
├── Personal Information
├── Appointments
├── Visits
├── Vital Signs
├── Diagnoses
├── Prescriptions
├── Laboratory Results
├── Medical History
├── Allergies
└── Billing History

13. Emergency Workflow

Emergency Patient
↓
Emergency Registration
↓
Immediate Triage
↓
Vital Signs
↓
Emergency Doctor
↓
Diagnosis
↓
Treatment
↓
Need Lab?
↙ ↘
YES NO
↓ ↓
Lab Treatment
↓ ↓
Results ↓
└──────┬──────┘
↓
Patient Condition
↙ ↘

```
 STABLE     CRITICAL
   ↓          ↓
Discharge   Admission
```

14. Inpatient / Admission Workflow

```
Doctor recommends admission
        ↓
Admission Request
        ↓
Check Bed Availability
      ↙      ↘
  Available   Full
      ↓        ↓
Assign Bed    Waiting List
      ↓
Create Admission
      ↓
Nurse monitors patient
      ↓
Medication
      ↓
Lab Tests
      ↓
Doctor Reviews
      ↓
Patient ready for discharge?
      ↓
Generate Final Bill
      ↓
Discharge
```

15. Discharge Workflow

```
Doctor approves discharge
        ↓
Final diagnosis
        ↓
Discharge summary
        ↓
Final prescription
        ↓
Final bill
        ↓
Payment
        ↓
Discharge patient
        ↓
Follow-up appointment
        ↓
Patient notified
```

16. Admin Workflow

```
ADMIN
 │
 ├── Dashboard
 ├── Users
 │    ├── Doctors
 │    ├── Nurses
 │    ├── Receptionists
 │    ├── Pharmacists
 │    └── Lab Staff
 ├── Patients
 ├── Departments
 ├── Appointments
 ├── Pharmacy
 ├── Laboratory
```

```
├── Billing
├── Reports
├── Inventory
├── Notifications
└── Settings
```

17. Reports Workflow

```
Reports
  │
  ├── Patient Reports
  ├── Appointment Reports
  ├── Doctor Reports
  ├── Laboratory Reports
  ├── Pharmacy Reports
  ├── Revenue Reports
  ├── Billing Reports
  ├── Admission Reports
  └── Department Reports
```

18. Role & Permission Workflow

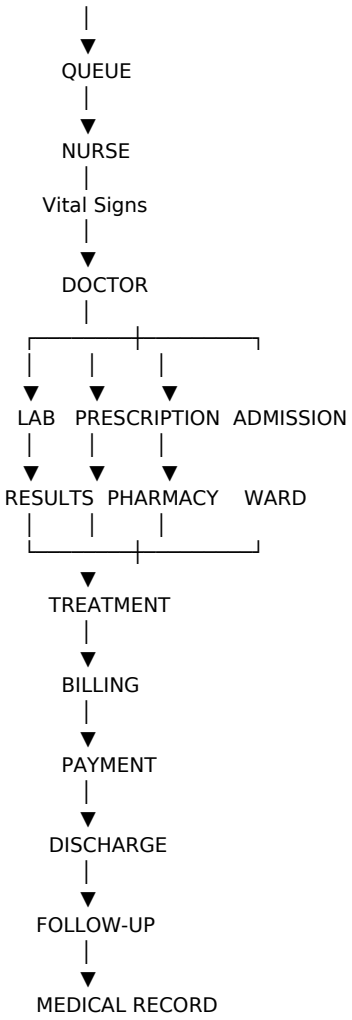
```
ADMIN
                  │
      ┌───────────┼──────────┐
      ↓           ↓          ↓
   DOCTOR       NURSE    RECEPTIONIST
     │            │          │
  Medical Data  Vital Signs  Registration
  Diagnosis     Nursing      Appointments
  Prescription  Notes        Billing
     │
     ├──── LAB
     │
     └──── PHARMACY
```

19. Patient Portal

```
LOGIN
  ↓
PATIENT DASHBOARD
  │
  ├── My Profile
  ├── My Appointments
  ├── Book Appointment
  ├── Doctors
  ├── Medical Records
  ├── Prescriptions
  ├── Laboratory Results
  ├── Bills
  ├── Payments
  ├── Notifications
  └── Logout
```

20. Complete End-to-End Hospital Flow

```
PATIENT
                  │
                  ▼
            REGISTRATION
                  │
                  ▼
             APPOINTMENT
                  │
                  ▼
              CHECK-IN
```



21. Recommended Figma Screens

AUTHENTICATION
- Login
- Forgot Password
- Reset Password

ADMIN
- Dashboard
- Patients
- Doctors
- Staff
- Departments
- Appointments
- Billing
- Reports
- Settings

DOCTOR
- Dashboard
- Patient Queue
- Consultation
- Medical Records
- Prescriptions
- Lab Requests

NURSE
- Dashboard
- Queue
- Vital Signs
- Nursing Notes

LABORATORY

• Dashboard
• Test Requests
• Sample Collection
• Results

PHARMACY
• Dashboard
• Prescriptions
• Inventory
• Dispensing

PATIENT
• Dashboard
• Appointments
• Medical Records
• Lab Results
• Prescriptions
• Bills
• Profile

22. Core Design / UX Requirements

The UI should prioritize:

1. Clarity
2. Usability
3. Consistency
4. Accessibility
5. Modern visual design
6. Real-world hospital workflow

Recommended common components:
• Sidebar navigation
• Top navigation
• Global search
• Breadcrumbs
• Cards
• Tables
• Filters
• Forms
• Modals
• Tabs
• Status badges
• Charts
• Notifications
• Toast messages
• Empty states
• Loading states
• Error states
• Confirmation dialogs
• Pagination

Project Development Order

For implementation, build the system in this order so the workflow remains manageable:

1. Authentication & Role Management
2. Hospital / Department Setup
3. Patient Registration
4. Appointment Management
5. Reception & Check-in
6. Nurse / Vital Signs
7. Doctor Consultation
8. Medical Records
9. Laboratory
10. Pharmacy & Inventory
11. Billing & Payments
12. Admission / Ward
13. Discharge & Follow-up
14. Reports & Analytics
15. Patient Portal
16. Notifications
17. Settings & Security
18. Testing & Final UI/UX Polish